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The Analysis of Neural Network Models to Distinguish AI generated faces from Real faces

Joshita Malla[1,*], Harshini Vemuri[1], SreeDivya Nagalli[1], S Abhishek[b], Anjali T[1]

[a]*Department of Computer Science and Engineering, Amrita School of Computing, Amrita Vishwa Vidyapeetham, Amritapuri, India*

[b]*Rolls-Royce Inc, Bangalore, India*

joshitamalla9@gmail.com

Abstract

The rise of Artificial Intelligence (AI)-generated faces that are identical to actual ones is both a technological innovation and a major concern. While considering the implications, some of the existing security systems are unable to differentiate between a high-quality deep-fake and an actual intruder's face. For instance, the risks are quite high at an airport security checkpoint, where facial recognition is the first line of security against unauthorized entry. The primary concern here is how trustworthy the computer programs and algorithms will be in recognizing counterfeits among a plethora of actual and artificially generated faces. This necessitates the need to introduce machine learning approaches to differentiate between actual and fraudulent faces, particularly when AI-generated faces are involved. Effective artificial intelligence systems must be adaptive and change quickly in the face of more complex threats rather than simply recognizing them. AI-generated faces are becoming more convincing by the day, increasing the risk of their exploitation. This poses the need to ensure that the technology on which people rely should be robust and trustworthy in critical situations, whether it is a security checkpoint or an e-commerce site. In this perspective, this study has attempted to develop and implement Artificial Intelligence-powered solutions to detect the artificial faces while ensuring reliability for critical functions in an age where reality constantly blends with fiction.

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Keywords:
Artificial Intelligence, Deep Fake, Deep Learning, Facial Recognition, Privacy, Security

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1. Introduction

In the era of rapid digital evolution, the line between natural and artificial has been increasingly blurred with growing technology. A typical day begins with social media feeds and shopping on e-commerce sites, and for many more things, people rely on facial recognition systems. IN such a case, people face a critical question as the facial recognition systems are currently used for security, browse through social media feeds, and engage with e-commerce platforms: Is it possible for current technology to quickly distinguish real faces and those created by machines? This research study aims to improve the safety and dependability of the technology that influences the world.

In this digital world where personal data is precious, it is necessary to make sure that the technology should have the ability to differentiate authentic and AI-generated faces effectively. This is an essential aspect to consider [1]. As the day passes, AI-generated faces are becoming more convincing, leading to an increase in the risk of their misuse. Its is highly recommended to ensure that the technology should be strong enough and equally reliable in critical situations, whether it be a security checkpoint or an e-commerce site. Deepfake has become a common issue in the current society, where people are morphing others and claiming the deepfaked one's to be the original and using the deepfaked images and videos for several purposes, which leads to threat.

This study aims to identify a clear differentiation between real and artificial faces. One of the revolutionary face-recognition technologies is Face recognition based attendance systems. A typical attendance system will be used to track and monitor the statistics of a student's track record. One of the challenges is to focus on differentiating between genuine, i.e., absolute and counterfeit face images.

Traditional attendance methods relied on physical measures like ID cards and biometric data like fingerprints, etc., making it easier and more convenient. Face recognition attendance systems have come into the image but also with challenging security aspects where the Authorizations must have specific anti-spoofing measures in order to detect the artificial and natural face images of a respective person.

The emergence of faces created by artificial intelligence that are identical to genuine ones is both a technological wonder and a serious concern. Think about the consequences if a security system is unable to discern between a high-quality deep-fake and an actual intruder's visage [2]. The stakes are very high at an airport security checkpoint, where facial recognition is the first line of defense against unauthorized entrance. A major worry is how trustworthy these systems will be in spotting fakers among a sea of real and artificially generated faces.

The HFDR, i.e., Human Face Detection and Recognition, is the technique of identifying an individual based on his visible traits, that is, their facial features inside an image. It's a verification method that uses facial feature analysis to verify an individual's identity.

In the era of rapid digital evolution, the line between natural and artificial has been increasingly blurred with growing technology. Our day begins with social media feeds and shopping on e-commerce sites, and for many more things, we rely on facial recognition systems. We face a critical question as we use facial recognition systems for security,

browse through our social media feeds, and engage with e-commerce platforms: Is it possible for current technology to quickly distinguish between real faces and those cleverly created by machines? Our research is aims on improving the safety and dependability of the technology that influences our world, and this question is at its core.

We are living in a world where personal data is precious, and we should make sure that the technology we use is able to distinguish the difference between authentic and AI-generated impersonators effectively. This is an essential aspect to consider [1]. As the day passes, AI-generated faces are becoming more convincing, leading to an increase in the risk of their misuse. We must ensure they the technology on which we depend is strong enough and equally reliable in critical situations, whether it be a security checkpoint or an e-commerce site. Deepfake has become a common issue in the current society where people are morphing others and claiming the deepfaked one's to be the original and using these deepfaked images and videos for several purposes which leads to threat.

We aim to keep a clear distinction between what's real and what's artificial. One of the revolutionary face-recognition technologies is Face face-recognition-based attendance systems way leading. A typical attendance system will be used to track and monitor the statistics of a student's track record. One of the challenges is to focus on differentiating between genuine, i.e., absolute and counterfeit face images.

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Our objective is to underscore the superior accuracy and reliability of the proposed HFDR using machine learning models and methods, which are effectively blended with human expertise and the computational power of artificial intelligence. The outcomes of the effectiveness of the respective approach will shed light on the comparison between natural and AI-generated face detection. eliability in the context of Human Face Detection and Recognition using machine learning models and AI means that the system consistently delivers accurate and trustworthy results across a range of scenarios and conditions, instilling confidence in its users.

2. Related Works

The study [3] by D. T. T Vijaya Kumar & R Mahammad Shafi, gives a novel suggestion is made to achieve instantaneous facial acknowledgement via custom-designed Fully Convolutional Network (R-FCN) mechanism that harnesses its position-aware pooling layer and by strengthening the prediction sectors related to Regions of Interest (RoIs).

Researchers have recently explored the avant-garde concept aimed at enhanced facial identification combining Grammatical Evolution (GE) with Grey Wolf Optimizer algorithm leading towards fast and precise face recognition to outperform other methods. While commendable in context speed and accuracy improvement targeting live time-face spotting processes, it signifies the substantial computational resource intake indicating the potential barriers, where working resources are limited.

The study [4] by Aswin Babu KV, Goutham Rajesh et al. explores the deployment of advanced facial identification software, intending to boost precision and simplify tallying presence. It highlights how such tech can reduce manual work required in logging participants at learning sites while detailing its numerous advantages. A probable hurdle could be moral dilemmas and confidentiality worries associated with using face-recognition tools, particularly within educational facilities. The study may not completely tackle all these nuances.

The study [5] by Bharath Tej Chinimilli et al. introduced an attendance system grounded on facials identification that employs the Haar cascade along with Local Binary Pattern Histogramn(LBPH) methodology. This system presents a secure as well as swift strategy for recording student presence. On account of central theory rooted around haar cascades combined implementation through LPBH algorithms possibly though one demerit concerning such setup remains sensitivity facing environmental also illumination situations which might magnify false negative cases.

The study [6] by T Anjali et al. delves into practical viability artificial intelligence(AI)'s application burgeoning field human face detection or identification. Authors are curious towards verification of exactness and efficiency regarding this method voicing heavy reliance artificial intelligence for human-face discernment.

Using simulation trials, the studies substantiate their proposition's superiority boosting the facial detection plus recognition prospects. However, a plausible weak-point could be AI-oriented module performance when negotiating broad range complexity amidst real-life situations. Results from modelled conditions validate its supremacy, practical implementation may face hurdles pertaining to discrepancies in light angles, person's posture along with faces expressing myriad emotions which might hinder desired accurateness.

In the research [7] done by RN Kulkarni, TSSL Anjali et al. The document showcases a technique to monitor student participation utilizing facial recognition tech and how it might fit into educational environments. It may delve into the operational implementation aspect of this system along with hardware and software prerequisites as well its inherent advantages for learning institutions. However one shortcoming could be challenges in managing lighting discrepancies or variations in pose or expression that can hinder face recognition systems' precision.

The work [8] by Kiruthika S. & Masilamani V. delves into the realm of image quality assessment for spotting counterfeit faces, often related to security and validation. It presumably examines an array of metrics linked with image quality as well as methods employed in distinguishing doctored or synthetic face images - a crucial measure safeguarding face recognition systems' integrity. This research may be handicapped due to its dependency on such metrics that might not remain solid against skilled falsification through cutting-edge technology. Moreover, it possibly overlooks the computational assets needed for immediate evaluation.

In comparison with the discussed concurrent studies, our research signals a notable breakthrough in facial recognition technology - more explicitly in differentiating between AI-produced and authentic human faces. Prior investigations into varying elements of face identification have stumbled across several limitations including an over-sensitivity to environmental changes, ethical matters or high computational resource demands.

Our study offers a stark contrast by deploying three robust deep learning models: VGG, EfficientNet, and ConvNeXt resulting in an outstanding 100 percent training accuracy and 99.9 percent validation accuracy success rate while separating AI-rendered from real faces. Of these algorithms employed, VGG emerges as the top-notch performer exemplifying that solving challenges posed by use of artificially produced content actually feasible via practical methods.

This revelation paints an optimistic future full of potential advancements enhancing both accuracy and operational efficiency for facial recognition techs through innovative algorithmic methodologies & architectures emanating eventually indomitable safety aspects related with using artificial creations responsibly within societal boundaries.

3. Dataset

The dataset utilized in this research consists of 140,000 images. These images are divided into two main groups: real and fake. Natural are images of real people's faces taken in real life. Fakes are images created by a computer to look like human faces. The images in the data set are stuck to a particular format, including image dimensions, color channels (e.g., RGB), and any pre-processing steps applied. Clarity on data format is fundamental for model compatibility. In the data set, 105,000 images are present in the train set, 21,000 images are present in the validation set, and 14,000 images are present in the test set. Every single image is of size 256x256 pixels.

The dataset consists of even a split of images in respective classes Fake-faces and Real-faces (i.e. 70,000 real and 70,000 fake). All images are in a certain way - like the right size and colors - for the computer to understand them. We also have special notes about where the faces are in the images. The images differ in many ways, just like in real life. Some have different lighting, backgrounds, or how the person's face looks. This is important because it helps our computer learn about all the different ways faces can look. The images in the data set are of high quality so that you can see the faces very clearly. This is important because it helps the computer do its job better.

4. Proposed Methodology

The main goal of this effort is to differentiate images of human faces from artificially generated images. This is attained using several models which results in high accuracy than other models. We made sure that we handle face images in a way that respects privacy regulations. It's also important to test the models resilience, against attacks and see how well it can handle possibly manipulated data. The techniques for detecting examples use innovative methods like federated learning to improve privacy preservation and adopt approaches that make model decisions more understandable and transparent.

4.1. ConvNeXt-T

ConvNeXt-T is an enhanced version of ConvNeXt, specifically designed for high temporal resolution tasks like video classification and action recognition. It enhances upon the original by incorporating a smaller patch size with many more layers. Its superb performance stems from coupling self-attention modules with numerous ConvNext blocks attaining state-of-the-art results in its field. The feature extraction and classification process involve training the custom ConvNeXt model, on a dataset comprising real and AI-generated face images. First, we get some images. The data preprocessing includes resizing, center cropping, and normalization. We let the CNN learn by sifting through these images. This is called "training." The adjustment of the learning process, however, is guided by a setting we call "ft."

The next step is categorizing them, this is done using layers tightly knit together in the model. We then test the model on other sets of images to see how good it is — this is what we call ”testing.” During this whole process, we keep track of things like accuracy, the amount of error (loss), and the F1 score. We do this for every round (epoch). At the end of each epoch, we save the model. The procedure goes on for several epochs. Experimental verification shows that the accuracy rate of the improved ConvNeXt[11].

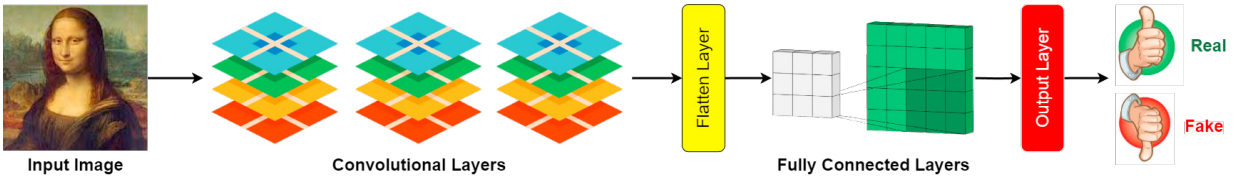


Fig. 1. ConvNeXt-T System Architecture

1) Patchify layer: This component transfers the input image into sequences of patches via a 4x4 convolutional layer that strides at 4.

2) ConvNexT stages: Four unique stages outline the architecture of ConvNext-T, each characterized by several ConvNext blocks followed by one downsampling layer. These self-attention modules to ensure efficient learning over long range dependencies in the input sequence.

3) Downsampling tiers: Reducing the size of sections - commonly known as pooling layers. These layers aim at cutting down on spatial dimensions, feature maps take up while preserving necessary details through processes such as max-pooling and average-pooling.

4) Classification head: This is a linear layer whose main function falls neatly under predicting outputs of the model. In various image categorizing works the ’classification head’ typically comprises one notch more densely packed layers.

Ultimately, optimized teaching approaches - such as data augmentation along label-smoothing, were used during training sessions strengthening this commendable architectural design. Thus enabling it act capable adeptness towards identifying longer-range sequencing patterns nestled quite deeply inside complex datasets being processed.

4.2. EfficentNetB3

EfficientNetB3 refers to an architecture for a convolutional neural network, which balancing the relationship between model scale and functionality. This network belongs to the EfficientNet group that is reputed for its heightened efficiency in contrast with numerous other deep learning patterns and keeps up with competitive accuracy as well. Its unique use of compound scaling permits adjustments on depth (layers), width (channels) and resolution(input image scale) altogether enhancing its accuracy-equilibrium-efficiency mix for this model type. Profiting from features such as Inverted Residual Block along with mobile-friendly operations like squeeze-and-excitation blocks plus depth-wise separable convolutions to capture efficiency of feature dependency. The model is a pretrained ImageNet feature extraction. A custom classification head is added to the bottom model, which involves batch normalization, dense layers and dropout to make it stronger in discriminative power. The dataset is divided into validation and test sets consisting of real and fake face images. In the training loop, for each epoch, progress details including loss and accuracy are continuously printed. It is provided with the user interaction that allows for manual interruption of training on prompt.

Learning rate adjustments by validation metrics enable flexibility of the model. This model surmounted the other models and conventional feature-based methods, achieving a remarkable accuracy[9][10].

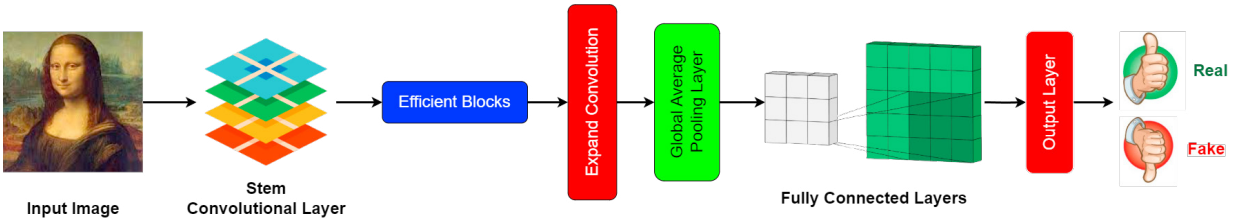


Fig. 2. EfficientNet B3 System Architecture

1) Convolutional Layers: In Convolutional Neural Networks (CNNs), The Convolutional Layers are of the picture details. Special filters, used by these layers, find small patterns like lines and textures in photos. This finds simple bits first, then builds the bigger picture. It's key for things like identifying what's in the picture. The deeper the layer, the better it can understand complexity.

2) Inverted Residual Blocks: The EfficientNetB3 model is filled with a succession of these units which bag and fine-tune features at varying scales. Their goal is to uplift the efficiency of neural networks by lessening parameters and computations but maintaining accuracy levels high.

3) Squeeze-and-Excitation Blocks: SE portions boost network's representational power by adjusting feature maps selectively for an enhanced focus on important characteristics. The "squeeze" phase includes global pooling that shrinks each feature map into one single value while set of weight learned during the "excitation" process are used for rescaling these maps.

4) Global Average Pooling: Global Average Pooling step used in exiting these convolutional layers but before entering fully connected ones - this method facilitates effectual summarization of key attributes from every map also depleting its spatial size. This mechanism turns 2D feature maps into a single dimension vector - this serves as precursor to fully connected layers.

5) Fully Connected Layers: This uses these globally averaged pooled layers, executing either classification or regression duties relying on specific application needs.

The design essentials like depth, size and aspiration in this model EfficientNetB3 secure optimal balancing accuracy and the model-actual size.

4.3. VGG16

In the realm of image classification and feature extraction, VGG16 stands out as a recognized convolutional neural network (CNN) outline. Totaling to 16 layers including three densely layers with thirteen convolution layers define its architecture. A complete pipeline for training a binary classifier to differentiate between real and AI generated face images using a VGG16-primarily based version. It starts with dataset organization and visualization, observed by means of data augmentation for the training set. The model structure, a modified VGG16, is defined with customizations to the final layer for binary classification. The training loop makes use of binary cross-entropy loss and Adam optimization, incorporating getting to know rate adjustments based on validation overall performance. Metrics along

with training and validation loss, accuracy, and confusion matrices are monitored and visualized over epochs. The VGG16 model presents the best accuracy compared to the other models for the case of image classification[12].

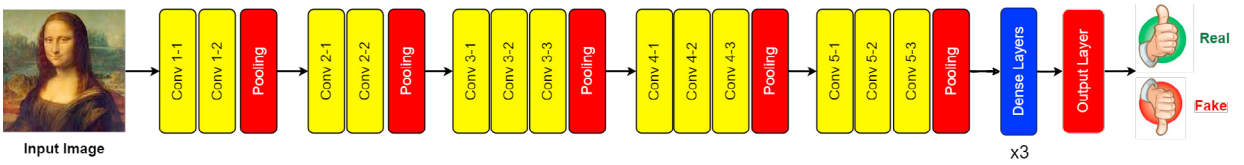


Fig. 3. VGG16 System Architecture

1) Input Layer: The beginning phase of a Convolutional Neural Network is known as the input layer, where it assimilates data. With this specific model, we're dealing with 224x224 pixel RGB imagery which enables color image processing.

2) Convolutional Layers: The convolutional layers are of fundamental significance in VGG16, where they employ small filters kits like 3×3 and 5×5 to extract features. The network can capture hierarchical, and complex visual patterns necessary for efficient image classification in these layers that systematically work on feed images.

3) Max-pooling Layers: Spatial downsizing happens via max-pooling layers that systematically reduce feature maps derived from their convolution layers. The most popular form combines usage of pooling windows measuring 2x2 pixels each.

4) Fully Connected Layers: The main strength lies within almost back-to-back connected subsequent sections carrying ReLU activations. The neurons intricately attached to all preceding-layer neurons.

5) Output Layer: It featuring neurons count ranging till 1000 paired alongside softmax activation for classification up until 1000 ImageNet classes.

5. Experiments and Results

The proposed research demonstrates their flexibility when dealing with challenging categorization problems. The results of this research provide further support for the conclusions and their possible practical applications, and they are a tribute to the careful effort and meticulous technique used in this research.

5.1. ConvNeXt-T

This analysis uses the ConvNeXt-T Convolutional Neural Network (CNN) model to tell real faces from those made by AI. The experiments involve training the model with varying data fractions, assessing its performance metrics such as accuracy and F1-score, and exploring different training configurations over 10 epochs. This model is great for visual tasks, making it perfect for our use. Good results were portrayed when running the analysis. During training we got 100% accuracy. Our model could tell the difference between real and AI faces with high accuracy. Validations tests showed it was almost perfect at 99.9%, showing it can really tell the difference that segregates the two. These results show this analysis works explicitly well and can easily tell real from fake.

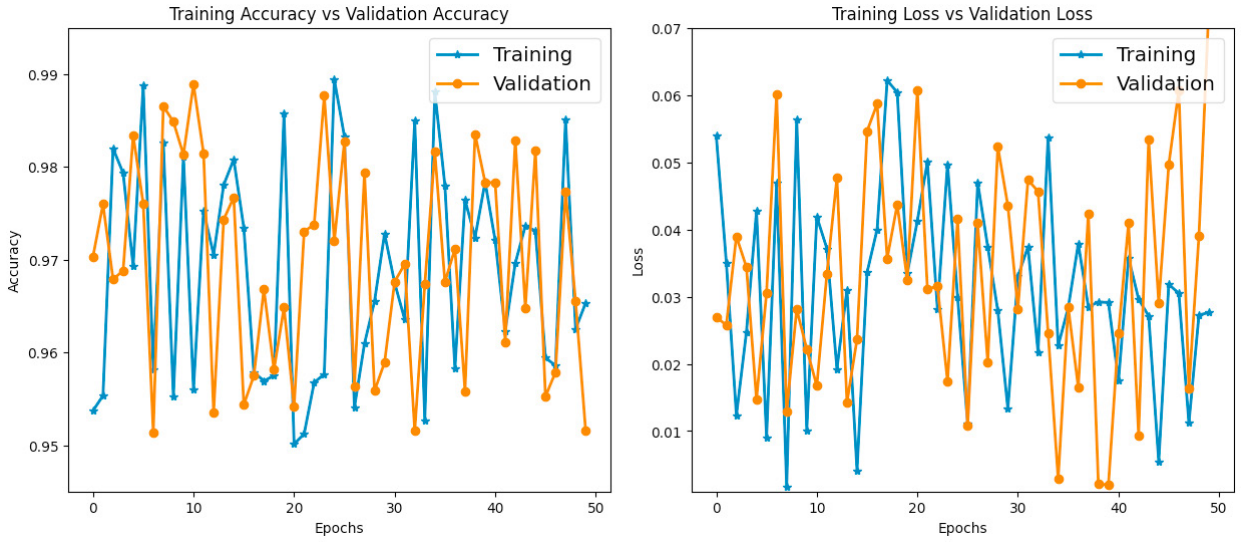


Fig. 4. Accuracy & Loss Curves of ConvNeXt-T

5.2. *EfficentNetB3*

This analysis uses a unique method based on the dependable cnn model EfficientNetB3 to address the given challenge. CNNs are made to distinguish or classify between images through distinctions. For this problem statement it is used to distinguish between real human faces and artificial intelligence-generated faces. In this experiment, the learning rates where adjusted dynamically according to the performance metrics. It was a good decision to put this strategy into practice due to its outstanding picture classification abilities, which were successfully balance model size and accuracy.

The result of this approach shows a training accuracy of 99.9%, validation accuracy of 99.9% and an F1 score of 0.9. This shows the competence of using EfficentNetB3 for analysis which remained to stay closely on par to the ConvNeXt-T model. This suggests that the EfficientNetB3-based method for comparing AI and real face analysis is dependable and effective. This validates its potential with distinction of images which can be used for further research in this domain.

5.3. *VGG16*

This analysis breaks down artificial intelligence (AI)-generated faces and human faces using the VGG16 Convolutional Neural Network (CNN) model. This model is an excellent fit for this analysis because of how well it can extract features from tasks that involves images. In this experiment, the Adam optimizer and binary cross-entropy loss is used as the optimization criterion . A learning rate scheduler is employed to dynamically adjust the learning rate during training based on validation performance. This analysis achieved 100% accuracy during training. Differentiating between faces with and without artificial intelligence was a high-accuracy task that was accomplished by our model. Testing its ability to distinguish between the two, it performed almost flawlessly in validation tests, scoring 99.9%.

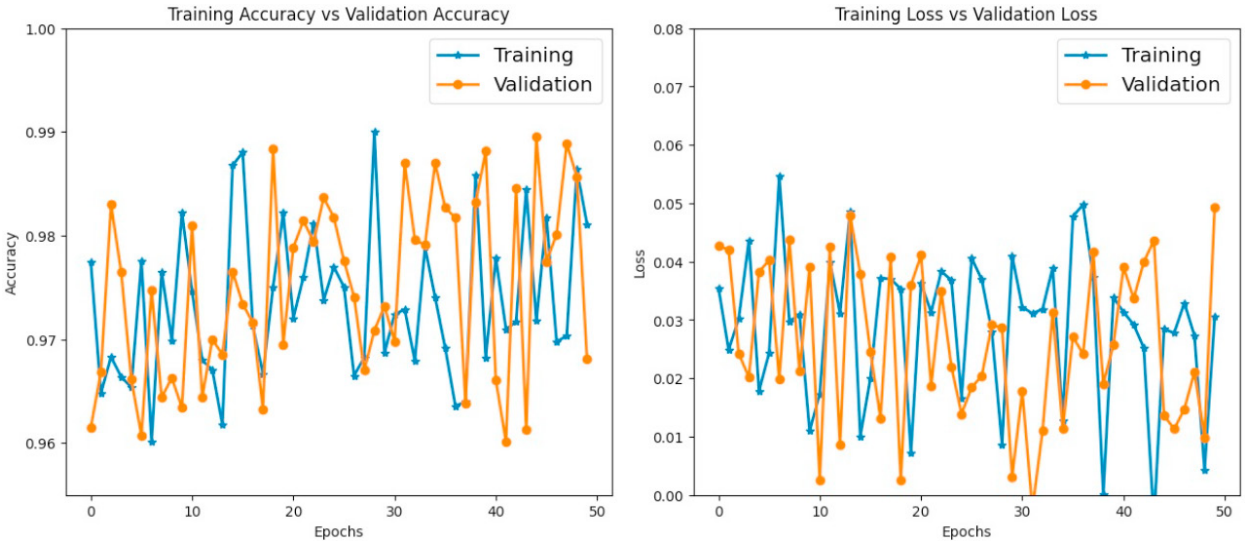


Fig. 5. Accuracy & Loss Curves of EfficentNetB3

The suitability of this approach in differentiating between the real and fake is demonstrated by these results. This shows great prospects for applications such as face recognition and deepfake detection as well as for future research works involving this area of study or similar application of models.

In this study, the use of three deep learning models—ConvNeXt-T, EfficientNetB3, and VGG16—was essential to obtaining outstanding outcomes and demonstrating their resilience in the study classification task. Strong evidence for the effectiveness of this research's methodology is provided by the three models' exceptional performance, high accuracy rates, and validation precision scores.



Fig. 6. Accuracy & Loss Curves of VGG16

6. Conclusion and Future Scope

This research study has primarily focused on classifying images to determine if an image is artificially AI generated or real. After using three distinct models namely VGG16, EfficentNetB3 , ConvNeXt-T we came to know that we got model accuracy hundred percent for VGG16 demonstrating it's effectiveness of determining the type of target image. As a part of future work we can take few factors into consideration like Further refinement and enhancement of models , using vast and different algorithms ,using large and more datasets and in order to test the robustness for further development of the models. In summary we have showcased the potential of three deep learning models in order to determine the type of the target image, i.e if the image is real or fake.

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